Climate Organizes but Does Not Determine Wildfire Development

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One-Sentence Summary

Climate shifts wildfire developmental opportunity without prescribing a single developmental path.

Abstract

Wildfire risk is commonly summarized by final burned area, duration, or spread rate, but fires can reach similar endpoints through distinct growth histories. We introduce Fire VASE, a developmental representation that converts daily FIRED events into comparable profiles and projects gridMET climate onto them. Across 278,569 U.S. events, 237,235 have complete centroid climate exposure from 2000 to 2021. Fire histories occupy recurring gradients of timing, persistence, pulse structure, reactivation, and termination. Hot, dry, low-fuel-moisture, and high-fire-danger conditions shift the prevalence of these forms, but blocked prediction and matched examples show that centroid climate does not uniquely determine them. Fire VASE reframes climate as shifting developmental opportunity: the growth histories a fire is more or less likely to follow.

Introduction

Climate is a first-order constraint on wildfire activity, especially through warming, drying, and rising atmospheric demand (1-3). Recent work has also sharpened attention on fire growth itself: the fastest-growing events account for disproportionate damage, and daily expansion can matter as much as final area for hazard and response (4). Yet continental analyses still commonly summarize fires by final burned area, duration, or average spread rate. Those summaries are indispensable, but they flatten development. A fire can reach the same final size through an early burst, steady accumulation, late surge, repeated pulses, or reactivation after quiescence.

Fire VASE was designed to preserve that missing developmental information. It maps developmental time to vertical position and cumulative burned area to ring width, producing a comparable object for every observed daily fire history. The underlying fire histories come from MODIS burned-area event delineation and FIRED perimeter products (5-7). Daily climate exposure comes from gridMET, a high-resolution gridded meteorological data set for ecological applications across the conterminous United States (8). Conceptually, Fire VASE draws from morphometrics, functional data analysis, and dimension reduction: it represents a history as a shape, compares shapes in a shared coordinate system, and then asks what external conditions shift the distribution of those shapes (9-11).

Here we ask how climate organizes wildfire developmental opportunity, defined as the distribution of growth histories made more or less likely under a given set of conditions. The paper makes two separable contributions. First, it defines a developmental representation that is estimated from fire histories before climate is considered. Second, it projects a comprehensive but centroid-based daily climate table onto that representation. The revised population table includes daily centroid maximum temperature, minimum temperature, vapor pressure deficit (VPD), wind speed, precipitation, relative humidity, specific humidity, 100-hour and 1000-hour fuel moisture, energy release component, burning index, reference evapotranspiration, potential evapotranspiration, and solar radiation for 237,235 climate-complete fires. Perimeter, active-burned-area, and perimeter-extension attribution remain a separate exposure product and are treated according to their actual coverage. The contribution is not a deterministic spread-rate predictor; it is a coordinate system for asking how climate shifts the probability of developmental forms and where centroid climate explanation reaches its limits.

Results

Fire VASE preserves developmental differences hidden by final outcomes

Figure 1 establishes the problem. Real fires with simple final summaries can have sharply different daily growth histories. Some accumulate most area early, others grow steadily, others surge late, and others develop through multiple pulses. The corresponding VASEs preserve these temporal differences in one visual grammar. This is the starting point for the climate analysis: climate should be evaluated against the whole developmental history, not only against final size or duration.

Wildfire histories vary along recurring developmental gradients

Figure 2 shows that observed fires occupy recurring developmental neighborhoods, but those neighborhoods lie along continuous gradients rather than forming isolated types. Labels such as skinny persistent, compact steady, late surge, front-loaded plateau, and multi-pulse complex are descriptive landmarks. The high prevalence of the single-flash neighborhood reflects the short duration of many mapped events, not a claim that most fires share a single mechanism. The quantitative result is a coordinate system for timing, persistence, concentration of growth, pulse structure, reactivation, and termination. Because this space is built from fire histories alone, climate can be projected afterward as an external correlate rather than baked into the axes.

Climate shifts the probability of developmental forms

Figure 3 projects climate onto Fire VASE space. Climate-colored morphospace maps show that maximum temperature, VPD, fuel moisture, and wind emphasize different portions of the developmental space, while composite VASEs show that low-, middle-, and high-VPD fires differ in where normalized growth is allocated through developmental time. Developmental-neighborhood prevalence shifts across VPD groups, and effect-size summaries show that maximum temperature, VPD, relative humidity, fuel moisture, precipitation, and fire-danger indices relate to different developmental responses. These associations are coherent with the broader literature linking warming, aridity, and fuel dryness to fire activity (1-3), but the VASE analysis resolves the outcome as a developmental distribution rather than a single aggregate burned-area response.

The predictive limit is equally important. In conservative blocked validation, which summarizes transfer across year, region, and region-year blocks rather than random-fire splits alone, the best transferable event-level representation is core event means, with median held-out R^2 of 0.349 across developmental responses. Region-season anomaly diagnostics do not outperform core event means, comprehensive event means do not outperform core event means, and temporally resolved exposure summaries do not outperform core event means in the median blocked comparison. This does not mean that humidity, fuel moisture, precipitation, or fire danger are unimportant. It means that, in these correlated daily centroid summaries and linear blocked baselines, adding more climate descriptors did not improve transfer beyond the core atmospheric variables. Modest blocked R^2 is therefore a bound on deterministic prediction, not a rejection of the distributional result. The claim is probabilistic: climate redistributes fires across developmental possibilities. It does not assign a unique developmental form.

Developmental state changes how climate is expressed through growth

The same daily climate exposure can occur before a fire begins rapid expansion, during the largest growth episode, or after growth has already tapered. We therefore modeled next-day growth as a function of climate, current developmental state, and their interaction. To avoid leakage, state was defined only from information available at day t: elapsed day, current daily growth, current cumulative area, and current acceleration. Final duration, final area fraction, and future VASE coordinates were not used.

State-containing models outperform climate-only baselines for next-day growth (Fig. 4). The best conservative state model is core climate-state interaction, with median held-out R^2 of 0.353. Core climate-state interactions survive the predeclared blocked-transfer margin. Because state predictors include current growth and acceleration, this gain is partly autoregressive. The result shows that recent fire state conditions the interpretation of near-term climate-growth associations; it does not show that the model has identified causal state-dependent climate control.

Climate organizes opportunity without uniquely determining outcome

Figure 5 asks where climate explanation fails. Pairs of fires with similar limited centroid summaries can have divergent VASE morphologies, and pairs with similar morphology can occur under contrasting climate pathways. These pairs are not matched on complete weather trajectories, active-edge exposure, or within-perimeter heterogeneity. That limitation is the point: mismatches define the scientific boundary of the current analysis. Climate describes opportunity. Which opportunity is realized likely also depends on active-edge exposure, local fuels, topography, vegetation, suppression, ignition context, human access, wind direction, and gusts. Prior work shows that human ignitions reshape the spatial and seasonal fire niche (12), active-fire studies show why daily fire progression can require spatially explicit growth tracking (13), and environmental controls such as fuels, vegetation, and topography shape fire occurrence, spread, boundaries, and transferability across landscapes (14-16). Those factors belong in the next version of the database, not in an overclaim from centroid climate alone.

Discussion

Fire VASE makes a common wildfire abstraction visible: final size is an endpoint, not a life history. Once the life history is represented directly, climate appears as a probabilistic shift in developmental opportunity. Hot, dry, high-VPD, low-fuel-moisture, and high-fire-danger conditions shift where fires fall in developmental space and when growth is allocated. But even expanded centroid climate does not collapse wildfire development into a deterministic sequence.

This framing changes how climate-fire relationships should be read. Event means are informative but blunt. Daily exposure, extreme-day fractions, and developmental timing sharpen interpretation, yet transfer across years and regions remains weak. Expanded centroid climate adds moisture, fuel, and fire-danger context, but it does not remove the need for spatially resolved exposure. Scaling perimeter and active-edge attribution, adding independent local climate normals, and integrating topography, vegetation, suppression, ignition context, wind direction, and gusts are the next necessary steps.

The present analyses are associational baselines. They do not isolate causal climate effects, suppression decisions, or fuel continuity. They also use daily centroid climate as the main population exposure, so they can miss within-perimeter heterogeneity, directional wind effects, and the climate experienced by newly burning edges. A stronger mechanistic account would need complete active-edge and newly burned-area climate, local climate normals, topography, vegetation, suppression, ignition context, wind direction, and gusts. Fire VASE supplies the coordinate system for that next layer of work: it shows that wildfire development occupies recurring forms, that climate shifts the probability of those forms, and that the realized path remains contingent on fire state and landscape context.

Materials and Methods

Fire histories were read from the repository's FIRED-derived daily VASE slice table, covering 278,569 events and 626,102 daily slices from 2 November 2000 to 1 May 2021. Climate exposure was read from the full-population climate-enhanced slice table. Complete daily centroid climate values were available for 237,235 fires. Variables were maximum temperature in degrees C, minimum temperature in degrees C, VPD in kPa, wind speed in m s-1, precipitation in mm d-1, maximum and minimum relative humidity in percent, specific humidity in kg kg-1, 100-hour and 1000-hour fuel moisture in percent, energy release component, burning index, reference evapotranspiration in mm d-1, potential evapotranspiration in mm d-1, and solar radiation in W m-2. Event-level climate summaries included means, daily minima and maxima, extreme-day fractions, early/middle/late developmental-time means, and a region-month fire-season anomaly diagnostic.

Developmental response variables were defined before model fitting and separated into absolute-scale outcomes, shape-normalized responses, and time-varying state variables. Event-level models used ridge-regularized linear baselines with fixed random seed 20260722. Predictors were standardized inside each training fold before fitting and then applied to the corresponding held-out fold. Validation used random fire splits as a diagnostic and year, region, and region-year blocking as conservative transfer tests; reported conservative R^2 values summarize the blocked tests. State-dependent models predicted next-day growth, $\log(1 + \mathrm{km}^2)$, using climate at day t and leakage-safe state variables available by day t. Because those state variables include current growth, cumulative area, and acceleration, these models are autoregressive associational baselines rather than causal estimates.

References and Notes

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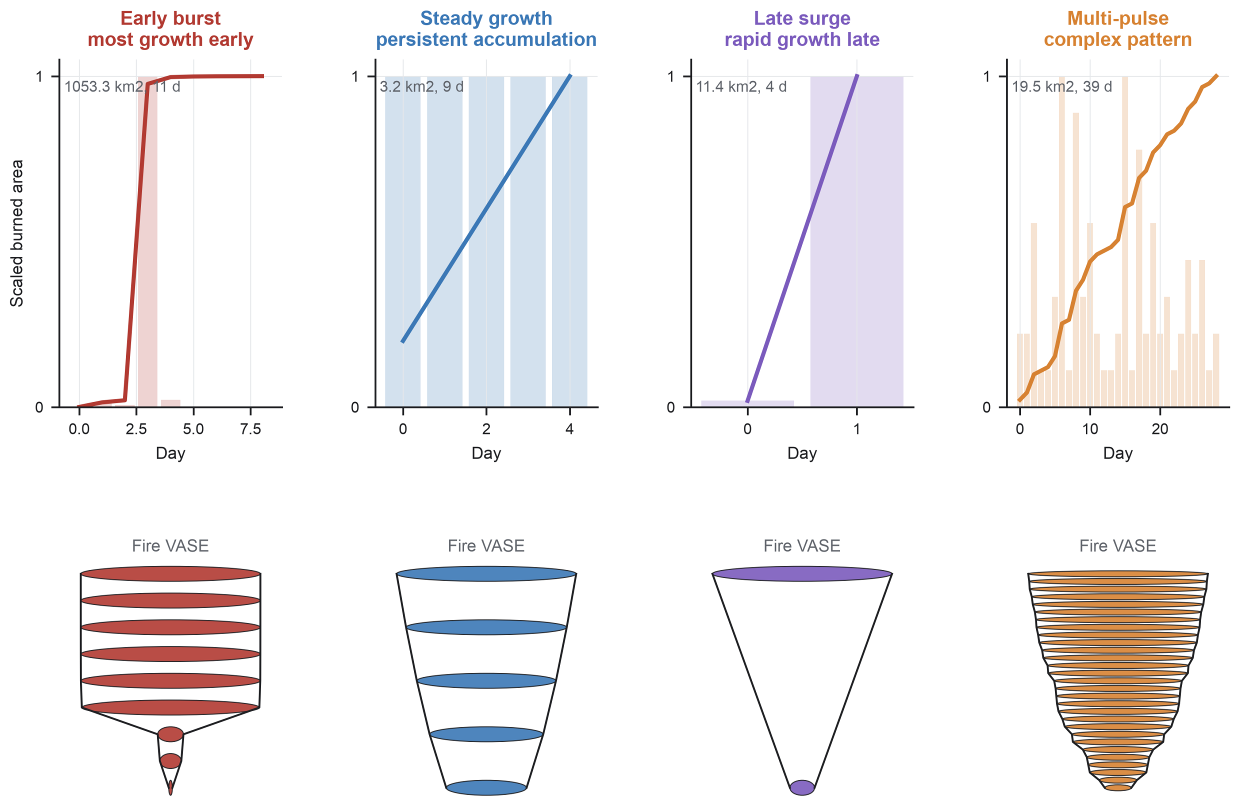
AI transparency: OpenAI Codex/ChatGPT was used as an AI-assisted coding, analysis, visualization, and editorial tool during development of this project. AI assistance included drafting and revising Python scripts for Fire VASE data ingestion, climate attribution, morphospace analysis, statistical summaries, figure generation, PDF/report production, and render-based quality checks; drafting and revising manuscript text, figure legends, response-to-review material, and simulated reviewer critiques; searching for and organizing candidate citations and author-guideline requirements; and helping maintain logs, manifests, tests, schemas, and documentation. The AI system did not originate the underlying FIRED, MODIS burned-area, gridMET, PRISM, or other observational data, did not make final scientific judgments independently, and is not listed as an author. Human investigators directed the analyses, selected the scientific claims, reviewed code and outputs, verified calculations and citations where reported, and remain responsible for the integrity, interpretation, and final content of the manuscript. Synthetic or illustrative demonstrations created during repository development are documented separately and were not used as evidentiary data for the manuscript analyses.

Supplementary Materials

Materials and Methods

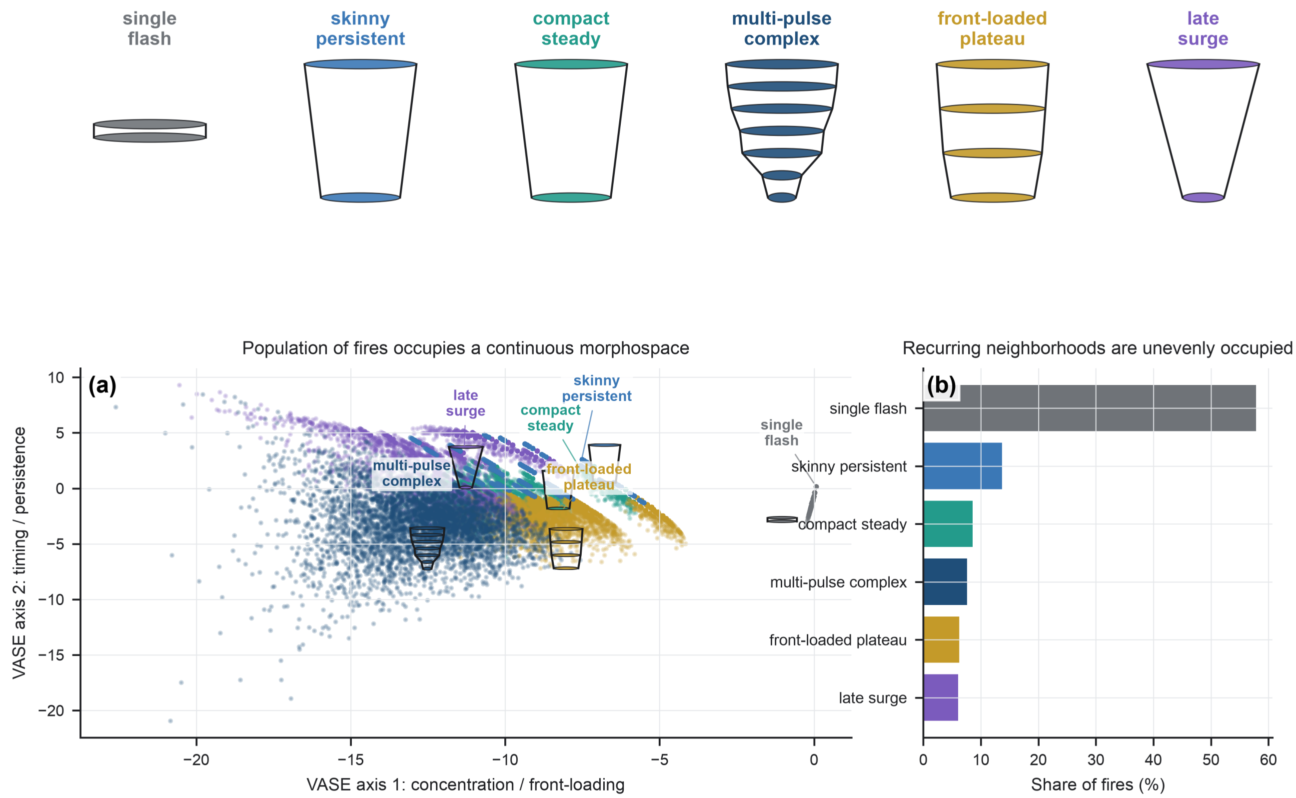
Figs. S1 to S3

Tables S1 to S4



Daily increments become ordered rings; ring width tracks cumulative burned area through developmental time.

Fig. 1. Fire VASE preserves developmental differences hidden by final outcomes. Four observed FIRED fire events show why final burned area and duration are insufficient descriptions of wildfire development. Each column follows one fire from its daily growth history to its Fire VASE representation. In the upper panels, bars show daily newly burned area and lines show cumulative burned area, both scaled within fire so that timing and sequence can be compared across events with different absolute sizes. In the lower panels, each daily history is encoded as a VASE: vertical position is ordered developmental time and ring width is cumulative burned area. The reader should see that fires with comparable final summaries can pass through different histories, including early burst, persistent accumulation, late surge, and multi-pulse growth. The scientific message is that wildfire development is a time-ordered life history, not only an endpoint; Fire VASE preserves the sequence of growth, pause, reactivation, and termination that final size and duration erase.



[x] Timing of growth

[x] Persistence

[x] Concentration of growth

[x] Pulse structure

[x] Reactivation

[x] Termination

Fig. 2. Wildfire histories vary along recurring developmental gradients. The population of observed fires occupies a continuous Fire VASE morphospace rather than a small set of isolated types. In (a), each point is one fire placed by the first two VASE axes, which summarize concentration or front-loading of growth and timing or persistence of growth. Point color marks the nearest descriptive developmental neighborhood, and representative VASE glyphs are overlaid in the morphospace so that the reader can see how morphology changes across the coordinate system. The labels are landmarks, not taxonomic classes: skinny persistent, compact steady, multi-pulse complex, front-loaded plateau, late surge, and single flash grade into one another. In (b), bars show the share of fires assigned to each neighborhood. The reader should see both recurrence and continuity: common forms exist, but their overlapping clouds imply developmental gradients. The scientific message is that Fire VASE provides a reproducible coordinate system for timing, persistence, concentration of growth, pulse structure, reactivation, and termination before climate information is added.

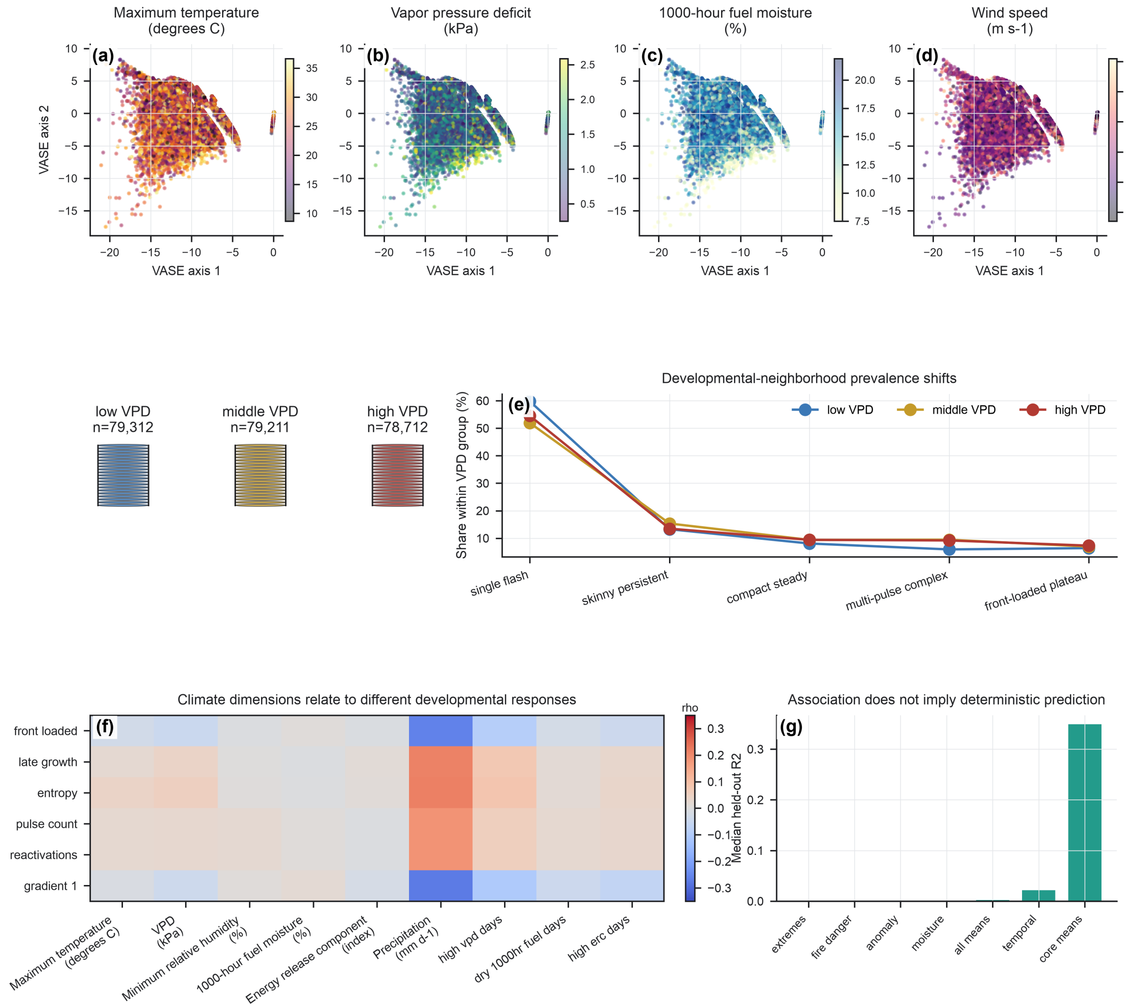


Fig. 3. Climate shifts the probability of developmental forms. Climate variables are projected onto the Fire VASE morphospace after the axes are estimated from fire histories alone. Panels (a) to (d) show the same fires in the same morphospace, colored separately by event-mean maximum temperature, vapor pressure deficit (VPD), 1000-hour fuel moisture, and wind speed from daily centroid gridMET summaries. These panels should be read as four climate lenses on one developmental space: warmer, drier, windier, and fuel-moisture conditions do not map to a single shape axis, but they emphasize different regions of the morphospace. Panel (e) asks whether this redistribution changes recognizable developmental neighborhoods by comparing low-, middle-, and high-VPD terciles; the adjacent composite VASEs summarize the median normalized growth profile and interquartile shell for each tercile, with group sizes in the titles. Panel (f) reports Spearman correlations between climate summaries and developmental responses, including front-loaded growth, late growth, entropy, pulse count, reactivation, and the first VASE gradient. Panel (g) shows conservative blocked validation for alternative climate representations; the best transferable baseline is core event means (median held-out $R^2 = 0.349$). The reader should see a distributional climate signal, not a deterministic classifier. The scientific message is that climate redistributes the probability of developmental forms while leaving substantial unexplained variation.

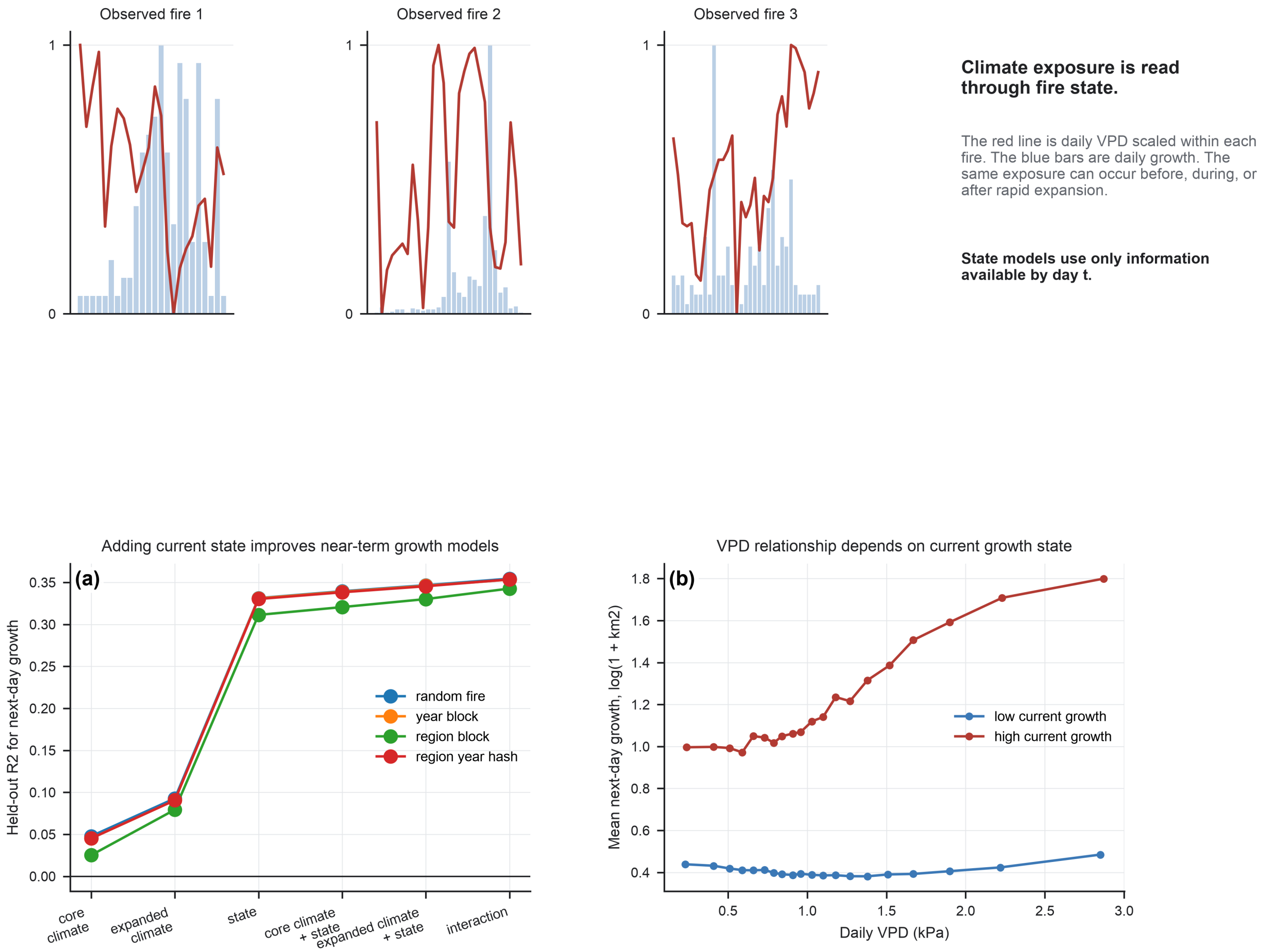


Fig. 4. Developmental state changes how climate is expressed through growth. The effect of a weather exposure depends on when it occurs in the developmental history of a fire. The upper row aligns daily growth with daily VPD for representative fires; values are scaled within each fire so that relative timing, not absolute magnitude, is emphasized. These examples show that similar VPD exposure can occur before, during, or after rapid expansion. In (a), leakage-safe next-day models predict log(1 + daily burned area in km^2) from core climate, expanded climate, current developmental state, climate plus state, and climate-state interactions. State predictors use only information available by day t, including elapsed day, current growth, cumulative burned area, and acceleration. In (b), conditional curves show that the empirical VPD relationship differs between low- and high-current-growth states. The reader should see that climate does not act on a blank slate: the same exposure can have different associations depending on the current trajectory. The scientific message is state dependence. Fire VASE turns climate-growth coupling from a static event summary into a developmental question.

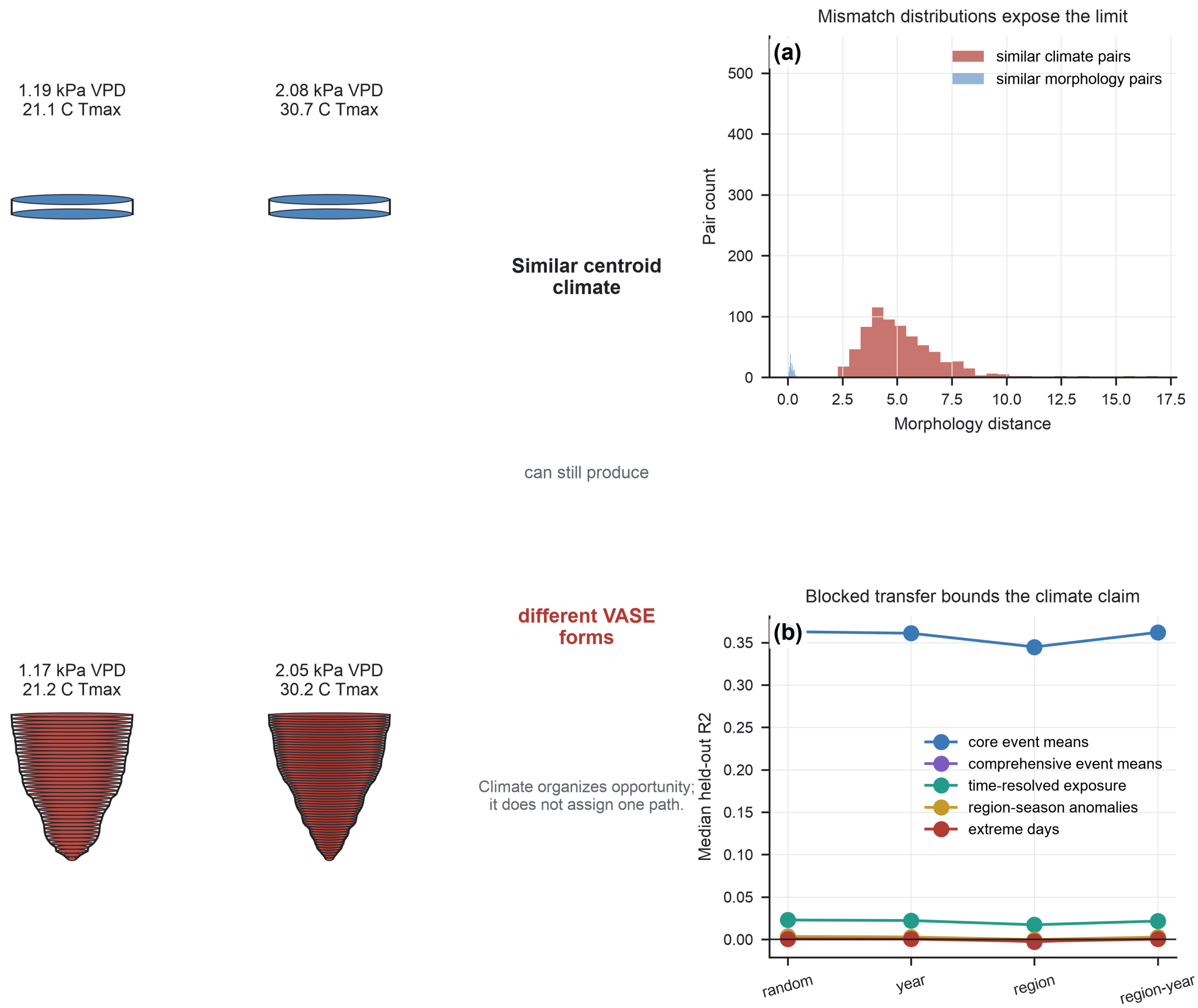


Fig. 5. Climate organizes opportunity without uniquely determining outcome. The final figure defines the boundary of the current climate interpretation. The VASE examples compare fires with similar limited centroid climate summaries but different developmental morphologies, and fires with similar morphology but different climate pathways. Labels give event-mean VPD and maximum temperature for the displayed examples. Panel (a) summarizes this mismatch at population scale by comparing developmental distances among pairs selected for similar climate and pairs selected for similar form. Panel (b) shows that climate-model performance declines under blocked validation across years and regions, consistent with the conclusion that centroid climate summaries organize opportunity but do not uniquely determine realized development. The reader should see why the residual variation is scientifically useful rather than merely inconvenient. The message is a limit and a roadmap: active-edge exposure, newly burned-area climate, within-perimeter heterogeneity, topography, vegetation, suppression, ignition context, wind direction, gusts, and local climate anomalies are the next data layers needed to explain which pathway a fire actually takes.